A New CPTG Gravitational Law

From Galaxies and Galaxy Clusters to Black Holes

Carter L Glass Jr

Abstract

This paper introduces a new gravitational law within Curvature Polarization Transport Gravity (CPTG): one continuous baryon-sourced source–polarization–transport law that can be applied quantitatively to individual galaxies, pressure-supported galaxy clusters, and finite black-hole states across radically different gravitational regimes. Rather than imposing one universal acceleration scale or one identical coordinate equation on every system, the law uses the measured baryonic distribution and physical state to determine the appropriate resolved or compact realization. Galaxies and clusters share the same resolved support-conditioned field relation, while black holes use the covariant finite mass-state realization of the same source–polarization–transport architecture.

Across the resolved galaxy and cluster tests, the conditioned CPTG closures reproduce the measured gravitational support with typical object-level median residuals of order one sigma or below, while localized larger residuals are retained and reported. The 153-galaxy SPARC calculation and the seven-cluster program use the same registered resolved field law, with X-COP and ACCEPT providing independent pressure-support reconstructions of the cluster states. Structural Mode N is then measured downstream from the solved fields as a separate readout of gravitational organization.

At compact-object field strength, the finite CPTG mass state closes analytically into source, transport, and polarization contributions and matches continuously to the exterior boundary, with the internal closure verified to machine precision. Two deliberately contrasting stellar-mass black holes are evaluated with the same compact relation without changing its form: the high-spin M33 X-7 branch gives a finite CPTG matching radius of 23.1099 km and $N_{\mathrm{BH}} = 1.49387$, while the low-spin LMC X-3 branch gives $r_m = 10.3072\,\mathrm{km}$ and $N_{\mathrm{BH}} = 1.83197$.

Together these results establish one CPTG gravitational architecture operating across galaxies, galaxy clusters, and finite black holes. To our knowledge, this is the first demonstrated baryon-sourced geometric framework to achieve quantitative closure across resolved galaxies, pressure-supported galaxy clusters, and finite black-hole states within one continuous source–polarization–transport gravitational construction, without assigning a separate invisible matter distribution to each system. The result shows that one gravitational law can accommodate measured structural diversity while preserving the appropriate physical coordinates of each regime.

1 Introduction

Gravity is observed through very different kinds of systems. In a spiral galaxy it is measured primarily through ordered rotation. In a pressure-supported galaxy or galaxy cluster it is reconstructed from the motions or thermal pressure required to support matter against collapse. Near a black hole, the problem becomes fully relativistic and is expressed through the mass, spin, curvature, and boundary structure of the compact state. A demanding test of a gravitational theory is therefore not whether it can describe one of these regimes in isolation, but whether the same physical law can remain coherent as the mass scale, geometry, support mechanism, and field strength change by many orders of magnitude.

Curvature Polarization Transport Gravity (CPTG) was developed as an extension of General Relativity rather than a replacement for it [1]. Einstein gravity remains the foundation of the theory and ordinary physical stress–energy remains the material source. CPTG adds two geometric response sectors. Curvature polarization describes how the gravitational field reorganizes in response to the strength, gradients, and structure of the baryonic source. Curvature transport describes how that organized response is retained, redistributed, or directionally displaced through an extended or evolving system. Where these additional responses become negligible, CPTG reduces continuously to the General Relativistic limit. The theory therefore does not switch from one law of gravity to another at a chosen acceleration or radius; it describes a continuous gravitational response whose additional geometric structure becomes important only where the physical state requires it.

That continuity does not require every object to have the same structure or to be represented by the same observational coordinates. The actual baryonic distribution, support state, geometry, and dynamical condition of an object must be measured. Here the *support state* means the gravitational acceleration required to sustain the observed dynamical state of the object: ordered rotation in a disk galaxy, stellar pressure in a dispersion-supported system, or thermal pressure in a hot cluster. It is an observationally reconstructed state coordinate, not an additional source of gravity.

A rotating galaxy, a pressure-supported cluster, and a compact black hole are therefore not forced into one artificial kinematic description. Instead, the appropriate observational interface is used to reconstruct the state that the common CPTG law must support. The CPTG structural acceleration $a_\star$ is the acceleration coordinate associated with the gravitational structure of the particular system being solved. It is not a universal acceleration constant; different CPTG reductions may determine the same structural quantity in different ways. In the resolved galaxy and cluster reduction used here, $a_\star$ is evaluated locally from the measured support state.

Structural Mode N is a separate CPTG quantity evaluated only after the gravitational solution is complete. It measures where the radial changes in the solved curvature-polarization/transport field are concentrated, and therefore describes the organization of the solved gravitational structure rather than the strength of gravity itself [4]. For a declared solved field, the resulting continuous N value is treated as a structural fingerprint of that realization. Independent reconstructions of the same object therefore retain their

own N values; they are compared directly and are never averaged, pooled, or replaced by a composite mode. It is not used to obtain the rotation curve or pressure-support solution. No dark-matter density profile is introduced as an additional material source.

The purpose of this paper is to test that cross-regime continuity as directly as possible. The registered support-conditioned CPTG relation is first applied to the primary SPARC galaxy sample, where the gravitational state is constrained by resolved rotation curves and the observed baryonic source [9]. The same resolved relation is then tested in pressure-supported systems, where the required gravitational field is reconstructed from Jeans or hydrostatic support. For the cluster program, the same seven objects are examined through two independent support reconstructions: the X-COP thermodynamic products and a separate ACCEPT-based reconstruction [11, 12, 8]. The test then moves to the compact regime, where the resolved weak-field coordinates are no longer appropriate and the same parent CPTG source–polarization–transport organization is continued through the finite covariant mass-state construction and its General Relativistic exterior handoff [2].

The novelty claim made here is deliberately specific. The paper does not claim that galaxies, clusters, or black holes have never been modeled separately. It tests whether one baryon-sourced CPTG gravitational architecture can remain quantitatively coherent across all three while preserving the different observational and relativistic coordinates required by each regime. To our knowledge, no prior framework has demonstrated that complete quantitative chain under one continuous source–polarization–transport construction without assigning a separate invisible matter distribution to the individual systems.

The standard used here is deliberately strict. Numerical equation closure, observational agreement, and post-solution structural readout are kept separate. The galaxy and cluster calculations must reproduce the measured support state without inserting an unseen matter distribution, while the compact construction must remain finite and close consistently to the exterior General Relativistic state. The claim being tested is therefore larger than the success of any one fit: it is whether one baryon-sourced CPTG gravitational architecture can remain quantitatively coherent from galaxies, through galaxy clusters, to black holes while preserving the measured individuality of each object. The sections that follow test that claim regime by regime and then compare the resulting continuity across the full macroscopic range.

2 Cross-Regime Physical Statement

This section states the single claim tested throughout the paper. Different systems may require different observational coordinates and different reduced solutions, but they must remain realizations of the same CPTG source–polarization–transport architecture. The point is to define what is allowed to change from object to object and what must remain physically continuous before the numerical tests begin.

CPTG does not require one numerical solution, one structural scale, or one coordinate representation for every mass-bearing object. The same source–structural-state–polarization–

transport ordering appears across systems whose measured structures and observational interfaces differ strongly. The parent CPTG field organization is

$$G_{\mu\nu} + \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{1}$$

with ordinary matter retained as the material source [1].

The common ordering is

$$\begin{aligned} \text{material source} &\longrightarrow \text{resolved or covariant structural state} \\ &\longrightarrow \text{curvature polarization} \longrightarrow \text{curvature transport} \\ &\longrightarrow \text{solved gravitational response} \longrightarrow \text{GR overlap.} \end{aligned} \tag{2}$$

The same ordering does not require one coordinate representation at every scale. For rotationally supported galaxies the present implementation is an effective radial reduction of an axisymmetric disk data set. For pressure-supported systems the spherical support field is reconstructed from Jeans or hydrostatic balance. Compact black holes require a relativistic mass-state continuation. The paper therefore distinguishes a common CPTG architecture from the more specific coordinate reductions used to evaluate it.

Ordinary baryons remain the material source throughout the gravitational applications considered here. CPTG does not infer a universal unseen mass profile and does not require one universal response scale to be assigned to every system. The measured baryonic distribution, support state, symmetry, gradients, and dynamical condition define the physical problem to be solved. The reduced quantities that characterize the resulting geometric response are therefore state coordinates of that solution, not additional matter components.

Throughout the paper, *GR overlap* means the state in which the additional CPTG polarization and transport response has become negligible and the solution continuously reduces to the ordinary General Relativistic response sourced by the same matter. It is a continuity limit of the same theory, not a switch to a different gravitational law.

Regime	Interface	CPTG representation	GR-overlap statement
Disk galaxy	Effective radial state from $V_{\mathrm{obs}}(r)$	Registered support-conditioned radial reduction	Resolved response tends to the ordinary baryon-sourced GR limit
Pressure-supported system	Jeans or hydrostatic $g_{\mathrm{sup}}(r)$	Spherical support-conditioned reduction	Resolved response tends to the ordinary baryon-sourced GR limit
Compact black hole	Spherical Misner–Sharp state $\mathcal{F} = m_{\mathrm{MS}}/M_\infty$	Covariant source–transport–polarization mass functionals plus mass–spin handoff	Local compact stress response vanishes at the GR/Kerr handoff

Table 1: Common CPTG architecture and the coordinate reduction used in each tested regime.

With the common claim and the regime map fixed, the next step is to identify the reduced CPTG ingredients that are carried into the resolved calculations.

3 Parent Reduced Structure

Before applying CPTG to data, this section separates the pieces inherited from the parent theory from the rules registered only for the reduced calculations used here. This keeps the theoretical hierarchy clear and prevents a working radial realization from being mistaken for a universal coordinate formula.

The status terms used below have specific meanings. *Parent inherited* means that the element comes from the common CPTG framework. *Registered* means that a reduced realization has been adopted and retained for the stated regime. *Closed* means that the mathematical reduction is complete within its declared geometry. *Conditional* means that the relation is valid only under the explicitly stated boundary or handoff condition.

The static reduced curvature carrier is

$$C_{\rm red} = g_j g_j + \kappa_{\rm red}^2 (\partial_i g_j)(\partial_i g_j), \tag{3}$$

with reduced polarization representative

$$\mathcal{L}_{\rm pol}^{\rm red} = \beta a_\star^{2/3} C_{\rm red}^{1/3}. \tag{4}$$

The directional transport representative is

$$\mathcal{L}_{\rm trans}^{\rm red} = -\frac{\lambda_t}{2} \left(S^\alpha \Sigma_{\rm red} \nabla_\alpha \Phi \right)^2 . \tag{5}$$

The cross-regime paper uses a strict authority ladder. The field-plus-gradient carrier and the polarization/transport sector ordering are inherited from the parent CPTG structure. The nonlinear suppression

$$\left[1 + (g/a_\star)^{123/50}\right]^{-11/50} \tag{6}$$

is a registered component of the resolved galaxy reduction and is retained in the present cluster reduction. The support-conditioned denominator is the registered reduced transport closure used by the resolved calculations. The compact spherical mass functionals are the closed compact-sector continuation. The minimum mass–spin handoff remains explicitly conditional on its stated handoff condition. These statuses are not interchangeable.

Element	Status in this paper	Authority
Field-plus-gradient carrier	parent inherited	reduced CPTG structure
$C^{1/3}$ polarization representative	parent inherited	reduced CPTG structure
Nonlinear $123/50, 11/50$ response	registered	galaxy / cluster reduction
Local $a_\star(r_i)$	registered resolved state rule	local resolved-state evaluation
$1 + c_C \Sigma_C^2$ denominator	registered reduced closure	resolved transport closure
Compact mass functionals	closed spherical continuation	Misner–Sharp inverse solution
Compact mass–spin handoff	conditional compact closure	spin-conserving handoff
Rotating compact Structural Mode	registered post-solution readout	Kerr-compatible shell registry

Table 2: Claim authority used throughout the manuscript.

For the resolved implementation, the bare geometric operator is evaluated from the support field before applying the local $a_\star^{2/3}$ weight. Thus the radial dependence of $a_\star(r)$ is not differentiated inside the bare operator. The common Euler–Lagrange factor multiplying the $C^{1/3}$ representative is absorbed into the effective reduced normalization A_C. This is the operational reduced law being tested here; the paper does not replace it with a different radial action by differentiating the registered local state weight.

These ingredients now define the reduced response to be tested. The next section combines them into the explicit support-conditioned field law used for both galaxies and pressure-supported systems.

4 Resolved Support-Conditioned Field Law

This section builds the radial field law used in the resolved tests. The measured support state supplies the local dynamical structure, the baryonic distribution supplies the material source, and the CPTG response determines whether the two can be closed consistently without adding an unseen matter component.

Before writing the equations, the CPTG-specific quantities can be read physically as follows. C_C combines the local support-field strength with how rapidly that field changes in space. The length scale κ_C controls how strongly those spatial gradients contribute. The operator $B_{\rm pol}$ converts that field-and-gradient structure into the local polarization response. The coefficient A_C sets the effective strength of that polarization response for the solved object state. The coordinate Σ_C^2 measures the local radial organization of the support field and enters the reduced transport response, whose effective strength is set by c_C. Finally, $\mathcal{S}$ suppresses the additional geometric response as the solved acceleration becomes large relative to the local structural acceleration $a_\star$. These are state and response coordinates of the reduced solution; they are not additional matter components.

For a genuinely spherical support field,

$$\mathbf{g}_{\rm sup}(r) = -g_{\rm sup}(r)\hat{\mathbf{r}}, \tag{7}$$

the contracted field gradient is

$$(\partial_i g_{{\rm sup},j})(\partial_i g_{{\rm sup},j}) = [g'_{\rm sup}(r)]^2 + 2\left[\frac{g_{\rm sup}(r)}{r}\right]^2. \tag{8}$$

The corresponding spherical reduced invariant is

$$C_C(r) = g_{\rm sup}^2(r) + \kappa_C^2\left\{[g'_{\rm sup}(r)]^2 + 2\left[\frac{g_{\rm sup}(r)}{r}\right]^2\right\}. \tag{9}$$

For spherical measure r^2dr, the bare response operator is

$$B_{\rm pol}(r) = \frac{1}{r^2}\left[(r^2+2\kappa_C^2)g_{\rm sup}C_C^{-2/3} - \frac{d}{dr}\left(r^2\kappa_C^2 g'_{\rm sup}C_C^{-2/3}\right)\right]. \tag{10}$$

For disk galaxies, Eqs. (8)–(10) are used as the registered effective radial operator acting on the resolved rotation-curve state. The paper does not claim that this radial operator is the full axisymmetric tensor contraction of a thin disk. The galaxy results below therefore test the registered radial disk reduction, while the cluster application is the direct spherical realization.

The structural acceleration is local,

$$a_\star(r_i) = g_{\rm sup}(r_i), \tag{11}$$

and is recalculated independently at every resolved radius. In the authority hierarchy of the parent theory, this equality is the registered realization of the structural acceleration quantity in the present resolved support-conditioned reduction; it is not promoted to a universal definition of $a_\star$ for every CPTG realization [1]. In the galaxy realization, each local $a_\star(r_i)$ remains calculated directly from the observed rotation speed; a separate differentiable, uncertainty-weighted support reconstruction supplies the radial derivatives needed by C_C, $B_{\rm pol}$, and Σ_C^2. The reconstructed profile does not replace the measured local $a_\star$. The support-state coordinate Σ_C^2 records how the radial support profile changes with radius. It is a compact measure of the local organization of the measured support field, not a density of unseen matter. It is

$$\Sigma_C^2(r) = \frac{1}{3}\left[1 - \frac{d\ln g_{\rm sup}(r)}{d\ln r}\right]. \tag{12}$$

For the differentiable support reconstruction, the numerical admissibility condition is

$$\Sigma_C^2 \geq 0. \tag{13}$$

This condition is not an arbitrary-geometry theorem. Where necessary, the reconstructed support profile is constrained to satisfy nonnegativity; Σ_C^2 itself is not clipped, and values above unity are retained.

The registered nonlinear response factor is

$$\mathcal{S}(r) = \left[1 + \left(\frac{g_{\rm CPTG}(r)}{a_\star(r)}\right)^{123/50}\right]^{-11/50}. \tag{14}$$

The resolved field law is

$$g_{\rm CPTG}(r) = \frac{g_b(r) + A_C\, a_\star^{2/3}(r) B_{\rm pol}(r)\mathcal{S}(r)}{1 + c_C\Sigma_C^2(r)}. \tag{15}$$

The field is implicit because $g_{\rm CPTG}$ appears in Eq. (14). The object-level quantities

$$\kappa_C, \qquad A_C, \qquad c_C \tag{16}$$

are reduced state/closure quantities. Numerically they are recovered by a bounded nonlinear support-state solve; the use of least-squares or root-finding is an implementation

method and does not turn them into universal parent constants. Dimensional consistency gives

$$[\kappa_C] = L, \qquad [A_C] = L^{2/3}T^{-4/3}, \qquad [c_C] = 1. \tag{17}$$

At every resolved point the implicit field equation satisfies

$$\mathcal{R}_i^{\rm field} = [1 + c_C\Sigma_{C,i}^2]g_{{\rm CPTG},i} - g_{b,i} - A_C a_{\star,i}^{2/3} B_{{\rm pol},i}\mathcal{S}_i = 0. \tag{18}$$

This equation residual is a solver-consistency diagnostic. It is distinct from observational support closure. In the resolved data applications, the support state itself is reconstructed from observations, and the object-level reduced quantities are recovered by minimizing the observational support residual. A *conditioned closure* therefore asks whether the CPTG law can consistently reproduce an observed state when that measured state is also used to determine the object's structural realization. This is different from a withheld-support forward test in which the target support data do not participate in construction of the tested state. Consequently the present galaxy and cluster tests are conditioned inverse closures, not source-only withheld-support forward tests.

4.1 Resolved GR-overlap boundary

This subsection checks that the resolved CPTG law returns continuously to the baryonic General Relativistic limit when the additional polarization and transport response becomes negligible. No separate switching scale is introduced.

The resolved GR-dominated overlap is the joint zero-response boundary

$$A_C \to 0, \qquad c_C \to 0, \tag{19}$$

for which

$$g_{\rm CPTG}(r) \to g_b(r). \tag{20}$$

This is a state-space boundary of the reduced law rather than an empirical acceleration threshold. It is the weak-field resolved representation of the parent overlap condition

$$\Pi_{\mu\nu}^{\rm pol} + \Theta_{\mu\nu}^{\rm trans} \longrightarrow 0, \tag{21}$$

for which the parent field equation returns to its Einstein form. CPTG therefore does not impose a universal acceleration at which a different gravitational law switches on.

4.2 Useful response limits

Two limiting behaviors are useful when reading the numerical results. One controls the high-acceleration suppression of the added response, and the other explains how a stable physical field can remain well defined even when individual reduced coordinates move onto an asymptotic numerical branch.

For $g_{\mathrm{CPTG}}/a_\star \gg 1$,

$$\mathcal{S} \sim \left(\frac{g_{\mathrm{CPTG}}}{a_\star}\right)^{-1353/2500}. \tag{22}$$

When the gradient term dominates the invariant,

$$C_C^{-2/3} \propto \kappa_C^{-4/3}, \qquad B_{\mathrm{pol}} \propto \kappa_C^{2/3}, \tag{23}$$

so a finite response may occupy the compensating branch

$$A_C \propto \kappa_C^{-2/3}. \tag{24}$$

The observable field may therefore be well constrained even when an individual reduced coordinate lies on an asymptotic numerical branch.

With the field law and its limiting behavior defined, the first observational test can be made in rotation-supported galaxies, where the support state is measured directly from the resolved rotation curve.

5 Rotationally Supported Galaxies

The galaxy test uses the simplest resolved observational interface in the paper: ordered rotation. The measured rotation curve defines the support state, while the observed gas, disk, and bulge components define the baryonic source. The purpose is to test whether the registered CPTG radial response closes those states across a large and structurally diverse sample.

For a disk galaxy the resolved observational state is

$$g_{\mathrm{sup}}(r_i) = \frac{V_{\mathrm{obs},i}^2}{r_i}, \qquad a_{\star,i} = g_{\mathrm{sup}}(r_i). \tag{25}$$

The present workbench uses the SPARC rotation-curve components directly [9]. The baryonic source used by the retained calculation is

$$V_b^2 = Q(V_{\mathrm{gas}}) + Q(V_{\mathrm{disk}}) + Q(V_{\mathrm{bulge}}), \qquad Q(V) = V|V|, \tag{26}$$

with

$$g_b = \frac{V_b^2}{r}. \tag{27}$$

The tabulated SPARC stellar velocity templates are used at their unit-Υ reference normalization; no additional fitted Υ_d or Υ_b rescaling enters the CPTG source construction.

The support profile entering C_C, B_{pol}, and Σ_C^2 is reconstructed from the observed rotation curve with an error-weighted radial spline. Its differentiated profile is constrained where necessary to satisfy $\Sigma_C^2 \geq 0$, without upper clipping or changing the directly measured

local $a_{\star,i}$. The same observed velocities then enter the support residual used to recover (κ_C, A_C, c_C). This makes the calculation an inverse conditioned closure. It tests whether the registered CPTG radial reduction can close the observed rotation state using the declared baryonic source; it is not a source-only reconstruction of $V_{\rm obs}(r)$ from baryons alone.

The current primary sample contains 153 galaxies and 3168 resolved radial points. The implicit equation residual closes with maximum relative value approximately

$$2.07 \times 10^{-16}, \tag{28}$$

which verifies the numerical implicit solve. The observational residual summary is

$$\chi^2_{\rm CPTG} = 16472.292025, \qquad n_{\rm pts} = 3168, \qquad \chi^2_{\rm CPTG}/n_{\rm pts} = 5.199587. \tag{29}$$

and the median of the 153 galaxy median absolute residuals is

$$0.469614\,\sigma. \tag{30}$$

There are 127 galaxies below 1σ, 144 below 2σ, and 151 below 3σ in median absolute residual. These σ-values use the tabulated point-velocity uncertainties; systematic inclination uncertainty is not included in those point errors and the quoted residuals should be read accordingly. The median-of-galaxy-medians and the aggregate point-level $\chi^2/n_{\rm pts}$ summarize different parts of the residual distribution: the former describes the typical galaxy, whereas the squared aggregate statistic is much more sensitive to a smaller number of large point residuals. Thus the sub-σ characteristic result refers to the typical galaxy median and does not imply that every resolved point lies within 1σ. MOND is referenced here only to make one conceptual distinction clear. In its original formulation, MOND introduces a universal acceleration scale a_0 that is taken to be the same across gravitating systems [10]. CPTG does not use a universal low-acceleration constant. Its structural acceleration $a_\star$ belongs to the gravitating system and, in the resolved reduction used here, varies locally with the measured support state of that baryonic structure, as expressed by Eq. (25). The contrast relevant to this paper is therefore fixed universal acceleration scale versus structure-dependent acceleration coordinate. No MOND residual benchmark, fit ranking, or theory-competition claim enters the cross-regime result presented here.

The galaxy branch audit also matters for interpretation. The locked galaxy search domain is

$$10^{-4} \le \kappa_C \le 10^{7}\ \text{kpc}, \qquad 10^{-30} \le A_C \le 10^{5}\ \text{m}^{2/3}\text{s}^{-4/3}, \qquad 10^{-10} \le c_C \le 10^{3}. \tag{31}$$

Some fitted galaxy coordinates approach numerical search boundaries, including κ_C and c_C; such behavior identifies numerical or asymptotic branches rather than precision measurements of individual reduced coordinates. The differentiable support reconstruction maintains $\Sigma_C^2 \ge 0$ without clipping the evaluated transport coordinate. In the current 3168-point primary sample, 16 radii have $\Sigma_C^2 > 1$, and all 16 values are retained in the field equation. The cross-regime claim rests on the solved field architecture and its observational closure.

5.1 Measured Structure and Post-Solution Structural Mode N

Once the galaxy field has been solved, a separate question can be asked: how is that solved response organized? This subsection separates the measured structure needed to define the galaxy from Structural Mode N, which is evaluated afterward and therefore does not participate in fitting the field.

The physical structure of an individual object is not inferred from its total baryonic mass alone. Its baryonic distribution and resolved dynamical state are measured, and those measurements define the structural state used by the CPTG reduction. Structural Mode N has a different role: it is a dimensionless CPTG measure of where the radial changes in the solved curvature field are concentrated [1, 4]. Lower N means that the important structural changes are distributed farther outward across the solved field, while higher N means that they are concentrated farther inward. It therefore measures the organization of the solved response, not the strength of gravity and not an observed morphology label. It is evaluated only after the CPTG field has been solved, is not an additional fit parameter, and is not used to obtain the rotation curve.

The common CPTG content of Structural Mode is

$$N = \frac{R}{\lambda}, \tag{32}$$

where R is the outer solved structural domain and λ is the characteristic organization length derived from that solved response. The meaning of N is therefore common across regimes, while the realization of λ follows the geometry of the system being solved rather than being copied unchanged from another regime.

For the registered galaxy readout, define

$$C_N(r) = g^2(r) + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr}\right)^2, \qquad \kappa_{\text{struct}} = 1 \text{ kpc}, \tag{33}$$

with

$$w_N(r) = \left|\frac{dC_N}{dr}\right|. \tag{34}$$

For outer solved radius R, the registered galaxy organization length is

$$\lambda_{N,\text{gal}} = \frac{\displaystyle\int_{R/5}^{R} r\, w_N(r)\, dr}{\displaystyle\int_{R/5}^{R} w_N(r)\, dr}, \qquad N_{\text{gal}} = \frac{R}{\lambda_{N,\text{gal}}}. \tag{35}$$

The continuous physical quantity is N. Because it records the organization of a particular solved field, each N value is retained as a distinct structural readout. If the same object is solved from an independent reconstruction, the second solution receives its own N readout. The two values may be compared directly, but no averaged Structural Mode is defined. The Curvature Structure Mode Index (CSMI) is the downstream galaxy catalog

layer that maps ranges of the galaxy N realization into descriptive structural classes. CSMI does not enter the gravitational solve and does not supply a morphology label to it. Conventional morphology is available only afterward as a validation coordinate. Thus the relevant inference order is

$$\begin{aligned}\text{measured baryonic and dynamical structure} &\longrightarrow \text{solved CPTG field} \\ &\longrightarrow N \longrightarrow \text{structural organization.}\end{aligned} \tag{36}$$

The present paper retains N as a post-solution structural readout. It is not inserted into the pressure-supported field law as a branch selector or an additional response coefficient.

The galaxy calculation therefore establishes both the resolved field closure and the downstream structural readout. The current 153-galaxy numerical reconstruction and worked examples are documented separately in [5]. The next test changes the support mechanism rather than the gravitational architecture.

6 Pressure-Supported Systems and Galaxy Clusters

The next test asks whether the same resolved CPTG response survives when ordered rotation is no longer the observable that supports the system. Pressure-supported galaxies use stellar motions and a Jeans reconstruction, while hot clusters use thermal pressure gradients and hydrostatic balance. These are different observational interfaces to the same gravitational problem, not different gravity laws [1, 6]. In each case the baryonic source remains material and the reconstructed support field is a dynamical coordinate rather than a new matter component.

For a pressure-supported system, a Jeans or hydrostatic reconstruction supplies

$$g_{\rm sup}(r), \qquad a_\star(r) = g_{\rm sup}(r). \tag{37}$$

The reconstruction constrains the gravitational field required by the observed support state; it is not an additional material source.

A spherical support-equivalent mass

$$M_{\rm sup}(r) = \frac{r^2 g_{\rm sup}(r)}{G} \tag{38}$$

gives

$$\frac{\rho_{\rm sup}}{\bar{\rho}_{\rm sup}} = \frac{1}{3}\left[2 + \frac{d\ln g_{\rm sup}}{d\ln r}\right], \tag{39}$$

and hence

$$\Sigma_C^2 = 1 - \frac{\rho_{\rm sup}}{\bar{\rho}_{\rm sup}} = \frac{1}{3}\left[1 - \frac{d\ln g_{\rm sup}}{d\ln r}\right]. \tag{40}$$

Thus the same registered support coordinate used in the radial galaxy reduction has a direct spherical pressure-support interpretation in the cluster case.

The cluster material source is constructed independently from the enclosed hot-gas and stellar baryonic masses on the retained radial source grid,

$$M_b(r) = M_{\rm gas}(r) + M_\star(r), \qquad g_b(r) = \frac{GM_b(r)}{r^2}. \tag{41}$$

This baryonic source grid is held fixed between the X-COP and ACCEPT tests. What changes in the second test is the pressure-supported field $g_{\rm sup}(r)$ and its covariance, not the material source. The two cluster calculations therefore test whether the same CPTG response remains stable under an independent reconstruction of the support state rather than under a changed baryonic mass model.

6.1 Original X-COP Seven-Cluster Support Closure

The first cluster test uses the X-COP thermodynamic products to reconstruct the gravitational field required by the observed gas support. This establishes the original seven-cluster pressure-support closure against which the later independent reconstruction can be compared.

The retained seven-cluster closure uses A2029, A1795, A2142, A2319, A644, A85, and ZW1215, drawn from the X-COP hydrostatic analyses [11, 12]. The numerical record is restricted to the stated domains below. The common inner $0.47R_{500}$ boundary records the retained closure domain of those six profiles; no additional universal physical meaning is assigned to that numerical cutoff in this paper. A2029 uses its separately retained 60–670 kpc domain.

Cluster	Domain	$n_{\rm pts}$	Median $\lvert\sigma\rvert$	Max. $\lvert\sigma\rvert$	Reduced χ^2
A2029	60–670 kpc	52	0.118	1.007	0.091
A1795	inner $0.47R_{500}$	41	1.540	4.370	3.649
A2142	inner $0.47R_{500}$	40	0.663	2.111	1.190
A2319	inner $0.47R_{500}$	40	0.098	0.236	0.015
A644	inner $0.47R_{500}$	40	0.118	0.566	0.028
A85	inner $0.47R_{500}$	41	1.076	3.527	2.031
ZW1215	inner $0.47R_{500}$	40	0.033	0.256	0.005

Table 3: Original X-COP support-state closure of the full implicit resolved field law over the retained seven-cluster domains.

The aggregate support-state statistic is

$$n_{\rm pts,tot} = 294, \qquad \chi^2_{\rm tot} = 266.114, \qquad \nu_{\rm tot} = 273, \qquad \chi^2_\nu = 0.975. \tag{42}$$

Five of seven clusters have median absolute residual below 1σ, all seven are below 2σ, and the median of the seven cluster medians is 0.118σ. The separate implicit field residual is of order 10^{-15} and verifies equation/root closure rather than observational agreement.

Cluster	κ_C (kpc)	A_C $(\mathrm{m\,s^{-2}})^{2/3}$	c_C	Numerical branch
A2029	51.126	1.899×10^{-2}	1.000×10^{5}	c_C upper boundary
A1795	1.000×10^{-4}	1.839×10^{-2}	1.000×10^{5}	κ_C lower, c_C upper
A2142	169.202	3.423×10^{-7}	0.882	finite
A2319	1310.850	6.932×10^{-8}	0.137	large-κ_C branch
A644	61.409	4.793×10^{-7}	0.858	finite
A85	5.418	1.546×10^{-2}	1.000×10^{5}	c_C upper boundary
ZW1215	48.077	4.931×10^{-7}	2.086	finite

Table 4: Reduced coordinates recovered in the original X-COP support closure and their numerical branch status.

The boundary entries are asymptotic branch selections, not precision measurements of the corresponding individual reduced coordinate. The scientific content of the present cluster test is the simultaneous support-conditioned field closure across the retained domains.

That first closure establishes the cluster result with one support reconstruction. The next subsection tests whether the same field law remains stable when the pressure-supported state is reconstructed independently from a different observational archive.

6.2 Independent ACCEPT Support Reconstruction

This second cluster test changes the support reconstruction while deliberately keeping the established baryonic source grid. The public ACCEPT Chandra profile archive supplies an independently reconstructed pressure-supported field, so the test asks whether the same CPTG response survives a different support data stream rather than reproducing only one X-COP realization [8]. It is therefore an independent support-side closure test, not an entirely independent reconstruction of both source and support.

The corrected 400-draw reconstruction propagates the pressure and density uncertainties into the radial support covariance, applies the declared covariance regularization and model-domain uncertainty, and evaluates the unchanged support-conditioned CPTG field law. The authoritative aggregate result is

$$n_{\mathrm{pts,ACCEPT}} = 238, \qquad \chi^2_{\mathrm{cov}} = 102.029497, \qquad \nu = 217, \qquad \chi^2_\nu = 0.470182. \tag{43}$$

The pooled median absolute diagonal residual is 0.632370σ, the maximum is 3.103030σ, and the pooled median absolute fractional CPTG–support residual is 4.646%. The implicit field equation closes separately with maximum absolute residual $2.03 \times 10^{-20}\ \mathrm{m\,s^{-2}}$. The low covariance χ^2_ν is interpreted as closure inside the adopted correlated and deliberately conservative reconstruction covariance, not as extraordinary measurement precision.

Cluster	$n_{\rm pts}$	Reduced $\chi^2_{\rm cov}$	Median $\lvert\Delta\rvert/\sigma$	Max. $\lvert\Delta\rvert/\sigma$
A85	22	0.0354	0.0709	0.2740
A644	37	0.2490	0.5348	1.1702
A1795	33	0.7835	0.6316	1.4468
A2029	46	0.4497	1.0412	1.4899
A2142	29	0.3749	0.6504	1.4156
A2319	33	1.2915	2.3240	3.1030
ZW1215	38	0.0445	0.1802	0.4287

Table 5: Corrected ACCEPT support-reconstruction closure under the declared total covariance. The baryonic source grid is retained from the established seven-cluster calculation; the pressure-supported field is independently reconstructed from ACCEPT profiles.

6.3 Seven-Cluster Structural Mode N

For a pressure-supported spherical cluster, Structural Mode is rederived from the cluster's own solved organization scale rather than by reusing the galaxy 1-kpc readout. The common dimensionless definition remains Eq. (32), but the spherical geometry supplies the organization functional. For the solved cluster field $g_{\rm CPTG}(r)$, define

$$C_{N,C}(r) = 3g^2_{\rm CPTG}(r) + \left[r\frac{dg_{\rm CPTG}}{dr}\right]^2, \tag{44}$$

and

$$w_{N,C}(r) = \left|\frac{dC_{N,C}}{dr}\right|. \tag{45}$$

The factor of three and the shell measure follow from the three-dimensional spherical organization of the radial field. The cluster organization length is therefore

$$\lambda_{N,C} = \frac{\displaystyle\int_{R/5}^{R} r^3 w_{N,C}(r)\,dr}{\displaystyle\int_{R/5}^{R} r^2 w_{N,C}(r)\,dr}, \qquad N_C = \frac{R}{\lambda_{N,C}}. \tag{46}$$

The retained numerical realization uses second-order gradients on the full nonuniform cluster grid, keeps the sampled points satisfying $r > R/5$, and evaluates the two shell integrals by trapezoidal integration. It does not use the galaxy five-point smoothing filter or the fixed $\kappa_{\rm struct} = 1$ kpc galaxy gradient scale. Thus the cluster readout preserves the physical meaning $N = R/\lambda$ while deriving λ from the solved spherical organization itself.

In this paper, the X-COP solution provides the primary Structural Mode value for each cluster. The ACCEPT solution is evaluated independently and reported beside it as a support-reconstruction sensitivity test. The two N values for a cluster remain separate readouts of their respective solved-field realizations. They are never averaged, pooled, interpolated into a composite mode, or used to retune either pressure-support solution.

Cluster	Primary $N_{\text{X-COP}}$	$\lambda_{C,\text{X-COP}}$ (kpc)	ACCEPT N_{ACCEPT}	$\lambda_{C,\text{ACCEPT}}$ (kpc)
A2029	1.521388	424.829	1.397476	349.863
A1795	1.566115	342.628	1.507837	245.288
A2142	1.328084	486.664	1.760504	220.087
A2319	1.350667	456.777	1.400538	333.231
A644	1.400836	401.293	1.120615	436.301
A85	1.533717	366.525	1.428954	162.552
ZW1215	1.358032	454.300	1.304734	430.850

Table 6: Post-solution spherical Structural Mode N for the seven cluster fields. The X-COP value is the primary structural readout used in this paper. The ACCEPT value is an independent support-reconstruction sensitivity readout and is retained separately. Differences between the X-COP and ACCEPT readouts reflect their distinct radial coverage and reconstructed thermodynamic/support-field shapes; because N depends on derivatives of the solved field and on the retained outer domain, the two measurements need not coincide even when both support reconstructions close. The corresponding spherical organization lengths are shown explicitly. Structural Mode values are never averaged.

The primary X-COP values span $N = 1.328084$–1.566115. A1795, A2319, and ZW1215 show especially close independent ACCEPT readouts, while A2142 and A644 show the largest reconstruction sensitivity. Those differences are preserved because Structural Mode is intended to retain the individual solved organization of each field rather than collapse independent realizations into a single statistic.

The reduced coordinates are state-conditioned and are therefore not expected to remain numerically identical when the reconstructed support state changes. The physical field closure is the primary comparison. In particular, several ACCEPT solutions occupy finite or asymptotic limits with negligible, finite, or strong transport; such limits are allowed states of the same reduced response rather than separate object classes.

A2319 requires additional interpretation because it is a dynamically disturbed merging cluster. It is the most stressed object in the spherical ACCEPT closure, but the parent CPTG hierarchy already distinguishes quasi-static radial transport from strongly anisotropic dynamical transport. In a merger, directional transport need not reduce completely to the radial structural remnant used by the coherent spherical calculation. The A2319 residual is therefore retained as a transition/stress result of the declared spherical reduction; the field law is not retuned to force a merging system to behave like a relaxed cluster.

The pressure-supported tests complete the resolved weak-field part of the paper. The next step moves into the strong-field regime, where the same parent theory must be written in covariant compact-state variables rather than forced through the radial support equation.

7 Covariant Strong-Field Realization in the Compact Regime

The compact test asks whether the same CPTG source–polarization–transport organization can remain finite when the problem becomes fully relativistic. The coordinate description must change here, so this section replaces the resolved support variables with a covariant mass-state construction while keeping the parent physical hierarchy unchanged.

The resolved weak-field radial law is not extrapolated through a black-hole interior. The compact regime is relativistic, and the retained compact solution is the covariant strong-field realization of the same parent source–polarization–transport architecture, expressed through the closed spherical mass-functional finite state [2]. The asymptotic GR/Kerr mass parameter is distinguished from the physical compact-source mass,

$$M_\infty = M_{\rm c} + M_{\rm geom} = A_{\rm pol} M_{\rm c}. \tag{47}$$

The geometric contribution is a gravitational mass-equivalent response, not an additional matter component. This distinction has an immediate physical consequence. An exterior observer measures the total gravitational charge M_∞, while the material compact source itself contains only $M_{\rm c}$. Whenever $M_{\rm geom} > 0$, CPTG therefore permits a smaller physical compact-source mass $M_{\rm c}$ to generate the same exterior Schwarzschild/Kerr gravitational mass parameter M_∞. The difference is carried by the completed geometric polarization response rather than by additional matter.

This mass economy is one of the broader consequences of the compact CPTG picture. It provides the same structural mechanism used in CPTG interpretations of quiet or failed stellar collapse, vanishing-star candidates, hidden compact remnants, and objects in or near the conventional lower mass gap: the gravitational mass inferred from the exterior field need not equal the amount of physical baryonic mass that completed the compact transition [3]. Those applications are not retested here; their relevance is that they follow from the same mass split being tested in this section.

For spherical symmetry, CPTG uses the Misner–Sharp mass as a covariant measure of the gravitational mass enclosed by a sphere. The normalized state $\mathcal{F}$ is the fraction of the final exterior gravitational mass already enclosed at a given radius. It provides a single finite interior coordinate on which the source, transport, and polarization mass channels can be organized. Define

$$\mathcal{F}[g] = \frac{m_{\rm MS}[g]}{M_\infty} = \frac{c^2 r}{2GM_\infty}\left(1 - D^a r D_a r\right), \tag{48}$$

with

$$\alpha_{\rm c} = \frac{M_{\rm c}}{M_\infty}, \qquad f_{\rm geom} = 1 - \alpha_{\rm c}. \tag{49}$$

The compact decomposition assigns distinct physical roles to the three channels. The source channel follows the physical compact matter. The transport channel represents the interior redistribution that is active during the transition. The polarization channel represents the geometric contribution that accumulates into the completed exterior gravitational charge. The minimum no-extra-shape decomposition is

$$\mathscr{M}_{\rm c}[g] = M_{\rm c}\mathcal{F}, \tag{50}$$

$$\mathscr{M}_{\rm tr}[g] = M_{\rm geom}\mathcal{F}(1 - \mathcal{F}), \tag{51}$$

$$\mathscr{M}_{\rm pol}[g] = M_{\rm geom}\mathcal{F}^2, \tag{52}$$

and closes identically,

$$\mathscr{M}_{\rm c} + \mathscr{M}_{\rm tr} + \mathscr{M}_{\rm pol} = M_\infty \mathcal{F} = m_{\rm MS}. \tag{53}$$

The minimum finite spherical profile is

$$F(x) = \frac{1}{8}(63x^5 - 90x^7 + 35x^9), \qquad x = \frac{r}{r_m}, \tag{54}$$

with

$$F(1) = 1, \qquad F'(1) = 0, \qquad F''(1) = 0. \tag{55}$$

These are matching data, not optional shape conditions. In the minimum spherical construction they make the mass function and its first two radial derivatives join the constant-mass General Relativistic exterior continuously, so no distributional surface layer is introduced at the matching boundary.

The compact channels are

$$H_{\rm c} = \alpha_{\rm c} F, \qquad H_{\rm tr} = f_{\rm geom} F(1 - F), \qquad H_{\rm pol} = f_{\rm geom} F^2, \tag{56}$$

with

$$H_{\rm c} + H_{\rm tr} + H_{\rm pol} = F \tag{57}$$

identically.

7.1 Compact GR-overlap boundary

The CPTG compact solution remains finite throughout the interior and must join smoothly to the General Relativistic exterior field at the matching boundary. This matching is a required closure condition of the compact construction. The General Relativistic exterior is retained, while the singular interior continuation of the classical black-hole solution is not part of the CPTG physical branch. This subsection checks that the local CPTG polarization and transport stresses vanish at the matching surface while the completed gravitational mass remains in the exterior charge.

For a spherical mass primitive $H(x)$, define

$$\epsilon_H = \epsilon_0 \frac{H'}{x^2}, \qquad p_{r,H} = -\epsilon_H, \qquad p_{t,H} = -\frac{\epsilon_0}{2x} H''. \tag{58}$$

Equation (55) gives

$$H'_{\rm tr}(1) = H''_{\rm tr}(1) = H'_{\rm pol}(1) = H''_{\rm pol}(1) = 0. \tag{59}$$

Thus the local transport and polarization stress generators vanish at the matching surface even when the completed geometric mass is finite,

$$\mathscr{M}_{\rm tr}(r_m) = 0, \qquad \mathscr{M}_{\rm pol}(r_m) = M_{\rm geom}. \tag{60}$$

The matched exterior is therefore the corresponding General Relativistic solution—Schwarzschild in the spherical realization and Kerr under the conditional rotating handoff—while retaining the completed interior geometric contribution in the conserved mass charge M_∞. Here *completed* means that the local CPTG polarization and transport stresses have gone to zero at the matching surface, while the integrated gravitational effect generated by that interior response has already been incorporated into the conserved exterior mass. Vanishing local response at the boundary therefore does not require $M_{\rm geom}$ to vanish.

The mass functional in this section is spherical. Rotation enters the present paper first through the separate mass–spin handoff below and then, downstream of the solved compact state, through the Kerr-compatible determinant measure used to register Structural Mode. Neither step replaces the spherical mass functional by a new fitted radial law, and the resulting post-solution axisymmetric readout should not be confused with a new full axisymmetric interior field solve.

With the finite spherical interior and its GR handoff established, the remaining step needed for an observational rotating benchmark is a minimal mapping between compact-source spin and the exterior mass–spin state.

8 Minimum Compact Mass–Spin Closure

This section introduces only the spin information needed to connect the finite spherical compact state to a rotating exterior benchmark. It does not construct a full rotating interior; it defines the minimum boundary handoff that can be tested with published mass and spin measurements. In this context, a *handoff* is the boundary mapping that connects the finite CPTG source state to the exterior Kerr mass and spin while preserving the conserved angular momentum. It is not a second gravitational law and not a full rotating-interior solution.

Let

$$\hat{a}_{\rm c} = \frac{cJ}{GM_{\rm c}^2}. \tag{61}$$

The minimum compact handoff condition is

$$r_m = \frac{GM_\infty}{c^2} = \frac{GM_{\rm c}}{c^2}\left(1 + \sqrt{1-\hat{a}_{\rm c}^2}\right), \tag{62}$$

which gives

$$A_{\rm pol} = 1 + \sqrt{1-\hat{a}_{\rm c}^2}, \qquad 1 \le A_{\rm pol} \le 2. \tag{63}$$

Here $A_{\rm pol} = M_\infty/M_{\rm c}$ is the compact polarization mass-amplification factor: it states how much larger the exterior gravitational mass parameter is than the physical compact-source mass in this conditional compact realization. It is unrelated to the resolved weak-field coefficient A_C.

Conservation of total angular momentum across the handoff gives

$$\hat{a}_\infty = \frac{\hat{a}_c}{A_{\rm pol}^2}, \tag{64}$$

and therefore

$$\hat{a}_\infty^2 A_{\rm pol}^3 + A_{\rm pol} - 2 = 0. \tag{65}$$

For $0 \le |\hat{a}_\infty| \le 1$, the cubic has one physical root on $1 \le A_{\rm pol} \le 2$. This is the minimum conditional mass–spin handoff used for the benchmark below. The paper makes no stronger arbitrary-geometry claim for Eq. (65).

That boundary relation is now ready for an observational check using a compact object with published dynamical mass and spin estimates.

9 Rotating Black-Hole Benchmark: M33 X-7

This section applies the compact handoff to M33 X-7. The purpose is not to refit the source or claim a uniquely measured spin, but to test whether a published mass–spin branch maps consistently through the CPTG finite-state closure and returns the same exterior boundary.

M33 X-7 is retained as a conditional rotating benchmark because published mass and continuum-spin estimates exist, but those inputs are model dependent. The original dynamical analysis reported

$$M_\infty = 15.65 \pm 1.45\, M_\odot \tag{66}$$

[13]. The 2008 continuum-fit spin value was corrected by the 2010 erratum to

$$\hat{a}_\infty = 0.84 \pm 0.05 \tag{67}$$

for the conventional vanishing-ISCO-stress continuum branch [14, 15]. A later spectroscopic analysis reported a lower black-hole mass near $11.4\, M_\odot$ and also favored a lower spin near 0.6 [19]. A 2025 continuum study showed that the same X-ray spectra admit a strong spin–plunging-region-stress degeneracy: the conventional high-spin branch remains near 0.84, while a low-spin branch can fit when intra-ISCO emission is retained [20]. The calculation below is therefore a conditional CPTG mass–spin mapping of the conventional reference branch, not a claim that M33 X-7 presently has a uniquely established spin and mass pair.

Using the conventional central reference pair

$$M_\infty = 15.65\, M_\odot, \qquad \hat{a}_\infty = 0.84, \tag{68}$$

Eq. (65) gives

$$A_{\rm pol} = 1.088925987. \tag{69}$$

Hence

$$M_{\rm c} = \frac{M_\infty}{A_{\rm pol}} = 14.3719593\, M_\odot, \tag{70}$$

$$M_{\rm geom} = M_\infty - M_{\rm c} = 1.2780407\, M_\odot, \tag{71}$$

with

$$f_{\rm geom} = 0.08166394, \qquad \hat{a}_{\rm c} = A_{\rm pol}^2 \hat{a}_\infty = 0.996038237. \tag{72}$$

The matching radius for that mass normalization is

$$r_m = \frac{GM_\infty}{c^2} = 23.1098807 \text{ km}. \tag{73}$$

The source-only Kerr handoff evaluated from $M_{\rm c}$ and $\hat{a}_{\rm c}$ returns the same radius to numerical precision.

The observationally inferred $15.65\, M_\odot$ quantity is the exterior dynamical gravitational mass. A conventional one-to-one reading identifies that exterior mass directly with the physical compact source. CPTG separates the two: for the retained central branch, the physical compact-source mass is $14.3719593\, M_\odot$, while $1.2780407\, M_\odot$ of the exterior gravitational charge is supplied by the completed geometric response. Thus

$$\frac{M_{\rm geom}}{M_\infty} = 0.08166394, \qquad \frac{M_\infty}{M_{\rm c}} - 1 = A_{\rm pol} - 1 = 0.088925987. \tag{74}$$

The geometric response therefore supplies 8.1664% of the measured exterior gravitational mass, while the exterior gravitational mass is 8.8926% larger than the physical compact-source mass.

Quantity	Conventional one-to-one reading	CPTG central finding
Measured exterior gravitational mass	$15.65\, M_\odot$	$15.65\, M_\odot$
Physical compact-source mass	identified with $15.65\, M_\odot$	$14.3719593\, M_\odot$
Geometric mass-equivalent response	none in the one-to-one identification	$1.2780407\, M_\odot$
Geometric share of exterior mass	—	8.1664%
Exterior mass enhancement	—	8.8926%

Table 7: Conventional one-to-one interpretation and CPTG decomposition of the retained M33 X-7 central branch. The measured exterior gravitational mass is unchanged; CPTG changes its physical decomposition into compact-source matter and completed geometric response.

9.1 Finite Compact Size and Rotating Structural Mode

The mass–spin closure fixes more than the exterior mass conversion. Within the declared finite CPTG compact realization it also fixes the physical compact-state boundary, while the solved rotating geometry supplies a downstream measure of how the interior response is organized. The two quantities answer different questions: r_m gives the radius at which the finite CPTG compact state hands off to the retained Kerr exterior, whereas Structural

Mode N_{BH} measures where the solved source–transport–polarization organization is concentrated inside that rotating compact geometry. The equal-measure shell registry used below is registered in this paper as the minimum axisymmetric post-solution realization of Structural Mode for the retained Kerr-compatible mass-function geometry; it introduces no fitted spin coefficient and is not asserted to be the unique covariant spatial-volume construction.

For the Kerr-compatible mass-function geometry, write

$$\Sigma = r^2 + a^2 \cos^2\theta, \qquad a = \hat{a}_\infty r_g, \qquad r_g = r_m, \tag{75}$$

so that the exact determinant measure is

$$\sqrt{-g} = \Sigma \sin\theta. \tag{76}$$

The determinant-based stationary radial–angular measure accumulated to radius r is therefore

$$\mathcal{V}(r) = \frac{4\pi}{3} r(r^2 + a^2), \tag{77}$$

which defines the registered equal-measure shell radius

$$\varrho(r, a) = \left[r(r^2 + a^2) \right]^{1/3}. \tag{78}$$

At the compact boundary,

$$\varrho_g = r_g (1 + \hat{a}_\infty^2)^{1/3}, \tag{79}$$

and the normalized rotating shell coordinate is

$$\xi(x, \hat{a}_\infty) = \left[\frac{x(x^2 + \hat{a}_\infty^2)}{1 + \hat{a}_\infty^2} \right]^{1/3}. \tag{80}$$

The finite radial mass profile already fixes

$$U(x) = \frac{F(x)}{x^2} = \frac{1}{8} \left(63x^3 - 90x^5 + 35x^7 \right), \tag{81}$$

and the compact structural invariant and weight are

$$\mathcal{C}(x) = U^2 + (U')^2 + 2 \left(\frac{U}{x} \right)^2, \qquad w(x) = \left| \mathcal{C}'(x) \right|. \tag{82}$$

The rotating post-solution Structural Mode is then

$$N_{\mathrm{BH}}(\hat{a}_\infty) = \frac{\displaystyle\int_0^1 w(x)\, dx}{\displaystyle\int_0^1 \xi(x, \hat{a}_\infty) w(x)\, dx}. \tag{83}$$

No fitted spin correction is introduced. Mass, r_g, and c cancel from Eq. (83); rotation changes only the geometric registration of the already solved radial structural transitions. The construction recovers the spherical benchmark continuously, $N_{\rm BH}(0) = 1.912233171442$, and decreases monotonically with spin magnitude in the retained Kerr-compatible registry. The spherical compact invariant and Kerr-compatible rotating geometry are inherited from the compact construction [2]; the determinant-based rotating shell registry and the resulting M33 X-7 mode are the newly registered post-solution realization evaluated here.

For the conventional M33 X-7 central branch, $\hat{a}_\infty = 0.84$, direct evaluation gives

$$N_{\rm BH} = 1.493871633444. \tag{84}$$

The corresponding spin-adjusted outer shell radius and organization length are

$$\varrho_g = 27.6115063 \text{ km}, \qquad \lambda_N = \frac{\varrho_g}{N_{\rm BH}} = 18.4831853 \text{ km}. \tag{85}$$

Together with Eq. (73), the retained CPTG branch therefore assigns the object a finite compact-state radius

$$r_m = 23.1098807 \text{ km}, \qquad 2r_m = 46.2197614 \text{ km}, \tag{86}$$

as well as the independent dimensionless organization readout $N_{\rm BH} = 1.49387$. Here r_m is the CPTG finite-interior matching radius, not a relabeling of the exterior Kerr event-horizon radius; ϱ_g is the equal-measure radius used only for the rotating structural registry. For comparison, the retained exterior mass–spin branch has

$$r_+^{\rm Kerr} = r_g \left(1 + \sqrt{1 - \hat{a}_\infty^2}\right) = 35.6490 \text{ km}, \tag{87}$$

so the finite CPTG handoff lies deeper than the retained outer horizon, $r_m < r_+^{\rm Kerr}$.

Using the quoted conventional spin interval gives

$$N_{\rm BH}(0.79) = 1.510886570748, \qquad N_{\rm BH}(0.89) = 1.478317125522, \tag{88}$$

so the retained branch may be summarized as

$$N_{\rm BH} = 1.4939^{+0.0170}_{-0.0156} \tag{89}$$

when only the quoted ± 0.05 spin interval is propagated through the structural registry. This is not a complete systematic uncertainty budget for the continuum-spin model.

The combined result maps the same observed black-hole branch to both a finite CPTG compact-state size and a quantitative post-solution organization coordinate. To our knowledge, no prior baryon-sourced geometric black-hole analysis has reported both a finite interior matching scale and an independent post-solution structural-organization coordinate for the same observed black-hole branch. The statement is framework-specific: conventional black-hole analyses already provide horizon and spin-dependent geometric

scales, while the new quantities here are the finite CPTG interior boundary together with Structural Mode of the solved compact response.

$\hat{a}_\infty$	$A_{\rm pol}$	$\hat{a}_{\rm c}$	$M_{\rm c}/M_\odot$	$M_{\rm geom}/M_\odot$
0.79	1.12095	0.99266	13.961	1.689
0.84	1.08893	0.99604	14.372	1.278
0.89	1.05907	0.99825	14.777	0.873

Table 8: Conditional CPTG closure of the erratum-corrected conventional M33 X-7 high-spin branch at the original central dynamical mass.

Because $A_{\rm pol}$ is determined by spin, a change in the adopted mass normalization scales the compact and geometric masses while preserving their fractions. For example, using the later $11.4\,M_\odot$ mass estimate with the same conventional $\hat{a}_\infty = 0.84$ branch gives

$$M_{\rm c} = 10.4690\,M_\odot, \qquad M_{\rm geom} = 0.9310\,M_\odot, \qquad r_m = 16.8340 \text{ km}. \tag{90}$$

This row is a sensitivity calculation only; the newer mass and conventional high-spin continuum branch are not asserted to constitute one uniquely measured joint posterior.

The finite spherical interior was evaluated at 1001 radial points using Eq. (54). The identity Eq. (57) closes with maximum numerical residual

$$1.11 \times 10^{-16}, \tag{91}$$

which is a numerical verification of the analytic mass decomposition. The compact boundary also satisfies Eq. (59). These are internal closure checks; the empirical content of the benchmark lies in how the observationally supplied mass–spin branch maps through the CPTG handoff.

The M33 X-7 calculation provides the high-spin compact benchmark. A complementary test should probe the opposite part of the same mass–spin branch without changing any compact coefficient or equation. LMC X-3 supplies that low-spin comparison.

10 Low-Spin Black-Hole Benchmark: LMC X-3

This section applies exactly the same compact handoff used for M33 X-7 to LMC X-3. The organization deliberately mirrors the preceding benchmark so that the two objects can be compared directly. No compact coefficient, mass profile, spin relation, or Structural Mode definition is changed between the high-spin and low-spin calculations.

LMC X-3 is especially useful because its black-hole mass is dynamically constrained while its spin lies near the low-spin end of the observed stellar-mass black-hole range. The optical dynamical analysis gives

$$M_\infty = 6.98 \pm 0.56\,M_\odot, \tag{92}$$

from the measured binary orbit and inclination [16]. Continuum fitting of more than one thousand thermal-state X-ray spectra gives

$$\hat{a}_\infty = 0.21^{+0.18}_{-0.22} \tag{93}$$

at 90% confidence [17]. A later IXPE spectropolarimetric analysis independently retained the low-spin interpretation: contemporaneous spectroscopy again gave $\hat{a}_\infty \simeq 0.2$, while the polarization-only analysis placed a 90% upper limit below 0.7 [18]. The calculation below therefore uses the established central dynamical mass and central low-spin continuum branch as a conditional observed state, not as a claim that all model dependence in the spin measurement has disappeared.

Using the central reference pair

$$M_\infty = 6.98\, M_\odot, \qquad \hat{a}_\infty = 0.21, \tag{94}$$

Eq. (65) gives

$$A_{\rm pol} = 1.759699717. \tag{95}$$

Hence

$$M_{\rm c} = \frac{M_\infty}{A_{\rm pol}} = 3.9665859\, M_\odot, \tag{96}$$

$$M_{\rm geom} = M_\infty - M_{\rm c} = 3.0134141\, M_\odot, \tag{97}$$

with

$$f_{\rm geom} = 0.431721225, \qquad \hat{a}_{\rm c} = A_{\rm pol}^2 \hat{a}_\infty = 0.650274050. \tag{98}$$

The matching radius for the central dynamical mass is

$$r_m = \frac{GM_\infty}{c^2} = 10.3071544 \text{ km}. \tag{99}$$

The source-only Kerr handoff evaluated from $M_{\rm c}$ and $\hat{a}_{\rm c}$ returns the same radius to numerical precision. The mass economy is much larger than in M33 X-7: 43.17% of the exterior gravitational mass is supplied by the completed geometric response in the central LMC X-3 branch, compared with 8.17% in the retained M33 X-7 branch. Equivalently, the exterior gravitational mass is 75.97% larger than the physical compact-source mass for the LMC X-3 central branch.

10.1 Finite Compact Size and Rotating Structural Mode

The size and Structural Mode calculation now follows exactly the same registered Kerr-compatible shell construction used above. For LMC X-3,

$$\Sigma = r^2 + a^2 \cos^2\theta, \qquad a = \hat{a}_\infty r_g, \qquad r_g = r_m, \tag{100}$$

with the same determinant measure $\sqrt{-g} = \Sigma \sin\theta$, determinant-based stationary measure

$$\mathcal{V}(r) = \frac{4\pi}{3} r(r^2 + a^2), \tag{101}$$

and registered equal-measure coordinate

$$\xi(x, \hat{a}_\infty) = \left[\frac{x(x^2 + \hat{a}_\infty^2)}{1 + \hat{a}_\infty^2} \right]^{1/3}. \tag{102}$$

The compact invariant $\mathcal{C}(x)$, structural weight $w(x)$, and finite profile $U(x)$ are unchanged from Eqs. (82) and (83). Therefore

$$N_{\mathrm{BH}}(\hat{a}_\infty) = \frac{\displaystyle\int_0^1 w(x)\, dx}{\displaystyle\int_0^1 \xi(x, \hat{a}_\infty) w(x)\, dx} \tag{103}$$

is evaluated with no new fit parameter.

For the central LMC X-3 spin $\hat{a}_\infty = 0.21$, direct evaluation gives

$$N_{\mathrm{BH}} = 1.831967029641. \tag{104}$$

The corresponding spin-adjusted outer shell radius and organization length are

$$\varrho_g = 10.4564954 \text{ km}, \qquad \lambda_N = \frac{\varrho_g}{N_{\mathrm{BH}}} = 5.7077967 \text{ km}. \tag{105}$$

Together with Eq. (99), the central LMC X-3 branch therefore gives

$$r_m = 10.3071544 \text{ km}, \qquad 2r_m = 20.6143089 \text{ km}, \tag{106}$$

and the independent organization readout $N_{\mathrm{BH}} = 1.83197$. As for M33 X-7, r_m is the finite CPTG interior matching radius rather than the exterior Kerr event-horizon radius, while ϱ_g is used only by the rotating Structural Mode registry. For the retained central exterior state,

$$r_+^{\mathrm{Kerr}} = r_g \left(1 + \sqrt{1 - \hat{a}_\infty^2} \right) = 20.3845 \text{ km}, \tag{107}$$

again giving $r_m < r_+^{\mathrm{Kerr}}$.

The published 90% continuum-spin interval extends from approximately -0.01 to 0.39. The scalar mass-amplification and Structural Mode registries depend on spin magnitude; the sign remains an orientation datum. Propagating only that quoted spin interval gives

$$N_{\mathrm{BH}}(-0.01) = N_{\mathrm{BH}}(0.01) = 1.912006296644, \qquad N_{\mathrm{BH}}(0.39) = 1.712144549971, \tag{108}$$

so the central branch may be summarized as

$$N_{\mathrm{BH}} = 1.8320^{+0.0800}_{-0.1198} \tag{109}$$

when only the published spin interval is propagated. This is not a complete systematic uncertainty budget for the continuum-spin model.

$\hat{a}_\infty$	$A_{\rm pol}$	$\hat{a}_{\rm c}$	$M_{\rm c}/M_\odot$	$M_{\rm geom}/M_\odot$
-0.01	1.99920	-0.03997	3.491	3.489
0.21	1.75970	0.65027	3.967	3.013
0.39	1.49340	0.86980	4.674	2.306

Table 9: Conditional CPTG closure of the published LMC X-3 low-spin branch at the central dynamical mass $M_\infty = 6.98\,M_\odot$. The outer rows propagate only the quoted 90% continuum-spin interval; the dynamical mass uncertainty is not folded into these rows.

At fixed central spin, the dynamical mass uncertainty simply scales the physical and geometric mass shares and the matching radius. Using $6.42\,M_\odot$ and $7.54\,M_\odot$, the two one-dimensional mass-sensitivity limits are

$$M_\infty = 6.42\,M_\odot: \qquad M_{\rm c} = 3.64835\,M_\odot, \quad M_{\rm geom} = 2.77165\,M_\odot, \quad r_m = 9.48022\ \text{km}, \tag{110}$$

$$M_\infty = 7.54\,M_\odot: \qquad M_{\rm c} = 4.28482\,M_\odot, \quad M_{\rm geom} = 3.25518\,M_\odot, \quad r_m = 11.13409\ \text{km}. \tag{111}$$

These rows are sensitivity calculations only; the quoted mass and spin confidence intervals are not treated as one joint posterior.

The contrast with M33 X-7 is the important test. The same compact equations map a rapidly rotating $15.65\,M_\odot$ exterior state to a comparatively small 8.17% geometric mass fraction and $N_{\rm BH} = 1.49387$, while the low-spin $6.98\,M_\odot$ LMC X-3 state produces a 43.17% geometric fraction and $N_{\rm BH} = 1.83197$. No theory parameter is changed between the objects. The difference is generated by their measured mass–spin states.

Object	$M_\infty/M_\odot$	$\hat{a}_\infty$	$A_{\rm pol}$	$M_{\rm c}/M_\odot$	$M_{\rm geom}/M_\odot$	r_m (km)	$N_{\rm BH}$
M33 X-7	15.65	0.84	1.08893	14.372	1.278	23.1099	1.49387
LMC X-3	6.98	0.21	1.75970	3.967	3.013	10.3072	1.83197

Table 10: Direct central-branch comparison of the two compact benchmarks. The same conditional CPTG mass–spin closure and the same rotating Structural Mode registry are used for both objects.

The finite dimensionless profile and source–transport–polarization identity are the same analytic compact solution used for M33 X-7. The LMC X-3 calculation therefore supplies an independent object-level test of the mass–spin mapping and rotating structural registry rather than a second fitted compact profile.

The galaxy, cluster, and two compact benchmarks have now been completed separately. The next section places them side by side to identify the common physical content that survived each change of scale, support, and coordinate representation.

11 Cross-Regime Synthesis

This section compares the three tested regimes directly. The aim is to separate what changed because the observations and geometry changed from what remained invariant because the underlying CPTG gravitational architecture was the same.

The continuity claim does not mean that one coordinate equation is forced onto every object. Galaxies and clusters provide the strongest direct cross-check because they use the same resolved support-conditioned field law while changing the observational support interface from rotation to pressure. The compact sector then tests whether the same parent source–polarization–transport architecture survives the transition to a finite relativistic state with a mandatory General Relativistic exterior match.

The galaxy, cluster, and compact calculations are different coordinate reductions of one source–polarization–transport architecture. For the resolved regimes,

$$g_b,\ g_{\rm sup} \longrightarrow g_{\rm CPTG} \tag{112}$$

through Eq. (15); rotation and pressure support change the observational construction of the state but not the registered radial response form. In the compact regime,

$$M_{\rm c},\ J,\ \mathcal{F} \longrightarrow M_\infty,\ \text{finite spherical interior},\ \text{GR/Kerr exterior} \tag{113}$$

through the compact mass functionals and the conditional mass–spin handoff. M33 X-7 and LMC X-3 then probe widely separated spin states of that same compact relation; each yields a finite matching radius and a rotating post-solution Structural Mode through the same registered Kerr-compatible shell measure.

The common content is therefore

$$\begin{aligned} \text{measured baryonic source and physical state} &\longrightarrow \text{polarization} \\ &\longrightarrow \text{transport} \\ &\longrightarrow \text{closed gravitational response.} \end{aligned} \tag{114}$$

Structural Mode N is evaluated downstream as the dimensionless coordinate that locates where changes in the solved field organization are concentrated. In galaxies and spherical clusters the registry is radial; in the retained rotating compact benchmark the same post-solution idea is carried through the Kerr-compatible equal-measure shell coordinate. CPTG does not require the unmeasured morphology or dynamical history of an individual object to be inferred from its total baryonic mass before the physical state is specified.

Three numerical statements must remain distinct. First, the pointwise implicit field residual tests whether the equation was solved correctly. Second, the observational support residual tests conditioned closure against measured data. Third, a source-only withheld-support forward test would require the target support field not to participate in construction or recovery of the tested state. The present galaxy and cluster calculations use the measured dynamical state of the object and therefore belong to the conditioned-support category; this is distinct from adding an unseen matter distribution to the source.

The compact mass identities are analytic closure relations evaluated on observationally supplied mass–spin inputs. Structural Mode N, where evaluated, is downstream of the field solve and tests organization that is not supplied as a morphology label.

That synthesis establishes the continuity claim. The next section states the exact boundary of that claim so the demonstrated result is not confused with stronger assertions that were not tested here.

12 Scope and Claim Boundary

This section states exactly what the calculations establish and what they do not. The purpose is to preserve the strength of the cross-regime result without extending it beyond the geometries, support reconstructions, and compact handoff that were actually tested.

The cross-regime result is limited only by the reduction actually used in each application.

- The SPARC calculation tests the registered effective radial disk reduction, not a newly claimed full axisymmetric derivation. Its current 153-galaxy statistics correspond to the direct tabulated gas, disk, and bulge source construction and to the tabulated point-velocity uncertainties.
- The seven-cluster program contains two support reconstructions. The original X-COP calculation tests the spherical pressure-supported reduction over the retained radial domains in Table 3; the ACCEPT calculation in Table 5 independently reconstructs the pressure-supported field while retaining the established baryonic source. Several cluster and galaxy reduced coordinates occupy numerical/asymptotic branches, so the solved fields are more sharply constrained than every individual reduced coordinate.
- The compact mass functional is the closed spherical finite continuation. Rotation enters through the conditional mass–spin handoff and, downstream, through the newly registered Kerr-compatible determinant-based shell registry used to evaluate the rotating Structural Mode. The M33 X-7 and LMC X-3 examples therefore provide conditional finite compact-state sizes and axisymmetric post-solution structural readouts, not new full axisymmetric interior field solves, unique covariant spatial-volume theorems, or unique empirical spin determinations.

No broader checklist of unfinished CPTG theory is required for the result of this paper. The claim made here is narrower and stronger: the registered CPTG source–polarization–transport architecture remains operational across individual galaxies, pressure-supported galaxy clusters, and the finite compact sector when each system is evaluated with the appropriate declared reduction.

With that claim boundary fixed, the conclusion can state the result without either weakening it or extending it beyond the evidence.

13 Conclusion

The conclusion brings the resolved galaxy, pressure-supported cluster, and compact-object results back to the single question posed at the start: whether one CPTG gravitational architecture remains quantitatively coherent across the macroscopic range while each object retains its measured structure and appropriate coordinate description.

CPTG is tested here as a macroscopic gravitational architecture rather than as a one-size-fits-all correction formula tied to one object class. Ordinary baryons remain the material source. The measured mass distribution, support, symmetry, gradients, and dynamical state of each object determine the appropriate reduction and the structural realization used by that solution.

In resolved rotation- and pressure-supported systems, the registered relation

$$g_{\mathrm{CPTG}} = \frac{g_b + A_C a_\star^{2/3} B_{\mathrm{pol}} \mathcal{S}}{1 + c_C \Sigma_C^2} \tag{115}$$

operates through different observational interfaces without introducing an unseen matter source. The 153-galaxy SPARC calculation tests the effective radial disk realization. Pressure-supported systems use Jeans or hydrostatic reconstruction of the field required by the observed state. At cluster scale, the original X-COP closure and the later ACCEPT support reconstruction provide two distinct pressure-support realizations of the same registered field law. The larger ACCEPT residual in the merging cluster A2319 is retained as a dynamical stress result consistent with the parent distinction between radial transport remnants and explicit anisotropic transport.

At compact scale, two contrasting stellar-mass black holes make the same continuity concrete in physical units. The high-spin M33 X-7 branch $M_\infty = 15.65\,M_\odot$, $\hat{a}_\infty = 0.84$ maps to $M_{\mathrm{c}} = 14.372\,M_\odot$, a completed geometric contribution of $1.278\,M_\odot$, $r_m = 23.1099\,\mathrm{km}$, and $N_{\mathrm{BH}} = 1.49387$. The low-spin LMC X-3 branch $M_\infty = 6.98\,M_\odot$, $\hat{a}_\infty = 0.21$ maps through the unchanged relation to $M_{\mathrm{c}} = 3.967\,M_\odot$, $M_{\mathrm{geom}} = 3.013\,M_\odot$, $r_m = 10.3072\,\mathrm{km}$, and $N_{\mathrm{BH}} = 1.83197$. The contrast demonstrates that the compact law is not being tuned to one black hole: different measured mass–spin states produce different finite mass decompositions and structural organizations while retaining the ordinary Kerr exterior.

The structure of an individual object is measured rather than guessed from its total baryonic mass. Structural Mode N is therefore downstream of the solution: it is the dimensionless CPTG measure of where radial changes in the solved curvature-polarization/transport field are concentrated. Table 6 extends that readout to all seven cluster solutions, retaining X-COP as the primary structural readout and ACCEPT as a separate reconstruction-sensitivity measurement. No Structural Mode values are averaged. This separates the measured physical state needed to define the problem from the theory's post-solution structural organization readout.

In the compact regime the coordinate representation changes because the problem is relativistic, not because the parent theory changes. The finite spherical Misner–Sharp mass functional preserves the source–polarization–transport organization, and the local

compact polarization and transport stress generators vanish at the GR/Kerr handoff while the completed gravitational charge remains M_∞. The conditional M33 X-7 and LMC X-3 calculations illustrate the associated mass–spin mapping at high and low spin and carry each solved state into the same Kerr-compatible Structural Mode registry, yielding finite matching scales and rotating organization readouts without promoting either model-dependent continuum spin estimate to a unique empirical state.

The cross-regime result is therefore a continuity statement: one CPTG source–polarization–transport architecture remains operational across macroscopic systems whose actual structures, support mechanisms, dynamical states, and strong-field coordinates differ. The GR limit is recovered continuously when the additional geometric response vanishes; no universal acceleration threshold or universal hidden-mass profile is required. Equation closure, conditioned observational closure, post-solution structural organization, and source-only withheld-support forward tests remain distinct evidentiary categories throughout.

The registered galaxy implementation used to produce the SPARC results is documented next; the publication evidence record follows immediately afterward so that the numerical claims remain directly traceable.

Data, Code, and Evidence Availability

The complete numerical evidence required to reconstruct and verify the solved results reported in this paper is collected in a single publication archive. The archive contains the authoritative source inputs, final solved numerical records, covariance data where applicable, and executable reconstruction and fail-closed verification code for the galaxy, X-COP cluster, ACCEPT cluster, M33 X-7, and LMC X-3 compact-object calculations. It is the publication evidence package for the quantitative results presented here.

`CPTG_ONE_GRAVITATIONAL_LAW_RECONSTRUCTION_EVIDENCE_20260927_FINAL.zip`

The publication evidence archive, together with supporting CPTG papers and numerical resources, is maintained in the public CPTG repository at:

`https://github.com/CLG2025/CPTG`.

14 Galaxy Analysis with the CPTG SPARC Browser Workbench

The CPTG SPARC Browser Workbench v1.13.1 is an interactive scientific application for calculating and examining galaxy rotation curves. Using published SPARC measurements and baryonic mass models, it analyzes individual galaxies or batches from the primary sample using the resolved CPTG procedure documented in the companion galaxy calculation paper [5].

For each galaxy, the interface displays the signed baryonic source, measured local structural acceleration, differentiated support, polarization, and transport alongside the fitted coefficients and solved velocities. Post-solution Structural Mode N, structural scale λ_{num}, and CSMI classification are reported separately, allowing the gravitational reconstruction to be examined at each radius.

The browser provides searchable galaxy selection, individual and batch execution, interactive rotation-curve plots, and detailed numerical results. Observed SPARC velocities appear alongside CPTG solutions and the familiar fixed-a_0 MOND reference. Separate velocity statistics include residuals, χ^2, and χ^2 per measured point for both calculations.

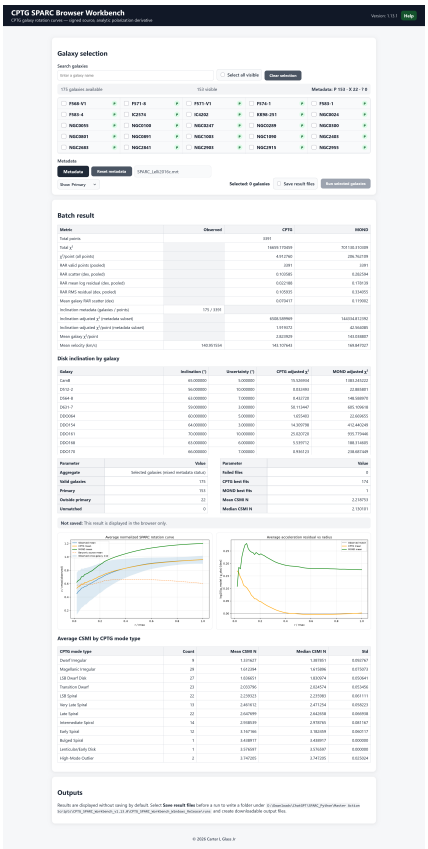


Figure 1: The CPTG SPARC Browser Workbench, showing batch-analysis results, sample statistics, and structural measurements.

The Radial Acceleration Relation (RAR) diagnostics are evaluated separately from velocity-based χ^2. At each valid radius, the observed acceleration $g_{\text{obs}} = V_{\text{obs}}^2/r$ is compared with each calculated acceleration. The logarithmic residual $\log_{10}(g_{\text{model}}/g_{\text{obs}})$ supplies the RAR scatter in dex, mean residual, RMS residual, and valid-point count for CPTG and MOND independently. These acceleration-space measurements do not alter either gravitational calculation.

Official SPARC metadata provide galaxy classification, distance, disk inclination, inclination uncertainty, and sample membership. Where available, inclination-adjusted velocity statistics are displayed separately without modifying the standard CPTG solution.

Batch summaries retain individual galaxy results while reporting successful and failed counts, primary-sample coverage, aggregate velocity χ^2, pooled RAR statistics, mean galaxy RAR scatter, and Structural Mode measurements. Full-precision radius-level exports, rotation-curve figures, and downloadable result packages support reproduction of the worked examples and further SPARC analysis [7].

References

[1] C. L. Glass Jr., *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, Zenodo (2026), DOI: 10.5281/zenodo.23288198.

[2] C. L. Glass Jr., *Curvature-Bounded Black Holes and Compact Objects in CPTG: Finite Interior Structure, Mass–Spin Closure, and Strong-Field Admissibility*, Zenodo (2026), DOI: 10.5281/zenodo.23172326.

[3] C. L. Glass Jr., *M31-2014-DS1 and the Vanishing-Star Problem in CPTG: A Finite Structural-Collapse Interpretation of a Contested Stellar Endpoint*, Zenodo (2026), DOI: 10.5281/zenodo.22431737.

[4] C. L. Glass Jr., *Structural Mode N in CPTG: Measuring Gravitational Organization from Galaxies to Black Holes*, Zenodo (2026), DOI: 10.5281/zenodo.23289859.

[5] C. L. Glass Jr., *Calculating CPTG Galaxy Rotation Curves: UGC 00731, NGC 7331, and NGC 2955*, Zenodo (2026), DOI: 10.5281/zenodo.23287928.

[6] C. L. Glass Jr., *CPTG Native-Phase Jeans and Structural Validation: XMM-VID1-2075*, Zenodo (2026), DOI: 10.5281/zenodo.22445804.

[7] *CPTG Supporting Python Models, Galaxy Calculations, and Research References*, CPTG companion repository.

[8] K. W. Cavagnolo, M. Donahue, G. M. Voit, and M. Sun, *Intracluster Medium Entropy Profiles for a Chandra Archival Sample of Galaxy Clusters*, Astrophys. J. Suppl. Ser. **182**, 12–32 (2009), DOI: 10.1088/0067-0049/182/1/12.

[9] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. **152**, 157 (2016), DOI: 10.3847/0004-6256/152/6/157.

[10] M. Milgrom, *A Modification of the Newtonian Dynamics as a Possible Alternative to the Hidden Mass Hypothesis*, Astrophys. J. **270**, 365–370 (1983), DOI: 10.1086/161130.

[11] S. Ettori et al., *Hydrostatic Mass Profiles in X-COP Galaxy Clusters*, Astron. Astrophys. **621**, A39 (2019), DOI: 10.1051/0004-6361/201833323.

[12] V. Ghirardini et al., *Universal Thermodynamic Properties of the Intracluster Medium over Two Decades in Radius in the X-COP Sample*, Astron. Astrophys. **621**, A41 (2019), DOI: 10.1051/0004-6361/201833325.

[13] J. A. Orosz et al., *A 15.65-Solar-Mass Black Hole in an Eclipsing Binary in the Nearby Spiral Galaxy M 33*, Nature **449**, 872–875 (2007), DOI: 10.1038/nature06218.

[14] J. Liu, J. E. McClintock, R. Narayan, S. W. Davis, and J. A. Orosz, *Precise Measurement of the Spin Parameter of the Stellar-Mass Black Hole M33 X-7*, Astrophys. J. Lett. **679**, L37–L40 (2008), DOI: 10.1086/588840.

[15] J. Liu, J. E. McClintock, R. Narayan, S. W. Davis, and J. A. Orosz, *Erratum: Precise Measurement of the Spin Parameter of the Stellar-Mass Black Hole M33 X-7*, Astrophys. J. Lett. **719**, L109 (2010), DOI: 10.1088/2041-8205/719/1/L109.

[16] J. A. Orosz et al., *The Mass of the Black Hole in LMC X-3*, Astrophys. J. **794**, 154 (2014), DOI: 10.1088/0004-637X/794/2/154.

[17] J. F. Steiner et al., *The Low-Spin Black Hole in LMC X-3*, Astrophys. J. Lett. **793**, L29 (2014), DOI: 10.1088/2041-8205/793/2/L29.

[18] J. Svoboda et al., *First X-Ray Polarization Measurement Confirms the Low Black Hole Spin in LMC X-3*, Astrophys. J. **960**, 3 (2024), DOI: 10.3847/1538-4357/ad0842.

[19] V. Ramachandran et al., *Phase-Resolved Spectroscopic Analysis of the Eclipsing Black Hole X-Ray Binary M33 X-7: System Properties, Accretion, and Evolution*, Astron. Astrophys. **667**, A77 (2022), DOI: 10.1051/0004-6361/202243683.

[20] A. Mummery, J. Jiang, A. Ingram, A. Fabian, and J. Rule, *A Rapid Black Hole Spin or Emission from the Plunging Region?*, Mon. Not. R. Astron. Soc. **544**, 2880–2896 (2025), DOI: 10.1093/mnras/staf1758.